=== UNIT 4: FUELS, CALORIFIC VALUES & COMBUSTION CALCULATIONS ===

>> 4.1 Classification and Calorific Value of Fuels

A fuel is a combustible carbonaceous substance that releases substantial thermal energy upon reaction with oxygen.
* Classification: Solid (Coal, Wood), Liquid (Petroleum, Diesel, Ethanol), Gaseous (Natural gas, LPG, Biogas).
* Calorific Value: Quantity of heat liberated by complete combustion of a unit quantity of fuel:
1. Higher / Gross Calorific Value (HCV / GCV): Total heat released when combustion products, including water vapor, are cooled to standard room temperature (15 deg C), capturing the latent heat of condensation of steam.
2. Lower / Net Calorific Value (LCV / NCV): Practical heat available when steam escapes with flue gases without condensing:
LCV = HCV - [Latent Heat of Steam * Mass of Hydrogen * 9]
LCV = HCV - [587 * (9 * H / 100)] kcal/kg = HCV - [52.83 * H] kcal/kg (or 2.45 MJ/kg in SI units).
* Dulong’s Empirical Formula: Estimates gross calorific value from elemental ultimate analysis:
HCV = (1 / 100) * [8,080 * C + 34,500 * (H - O / 8) + 2,240 * S] kcal/kg.
(O/8 accounts for combined hydrogen already bonded as water).

>> 4.2 Determination of Calorific Value via Bomb Calorimeter

* Apparatus: Heavy stainless steel cylindrical bomb lined with platinum/enamel. Weighed fuel (~0.8-1.0 g) in crucible ignited electrically via iron/fuse wire under 25-30 atm pure O2 pressure, immersed in weighed water (W) contained in copper calorimeter of water equivalent (w).
* Calculation Formula:
HCV = [(W + w) * (T2 - T1 + delta t) - (Cooling + Wire + Acid Corrections)] / Mass of Fuel (m).

>> 4.3 Coal Analysis: Proximate vs. Ultimate

* Proximate Analysis (Physical empirical testing):
1. Moisture (%): Heated at 105-110 deg C for 1 hour: % Moisture = (Loss in mass / Mass of coal) * 100.
2. Volatile Matter (%): Heated in covered crucible at 925 +/- 20 deg C for 7 minutes: % VM = (Loss in mass - Moisture) / Mass * 100.
3. Ash (%): Heated in open crucible at 700-750 deg C until constant weight: % Ash = (Mass of residue / Mass) * 100.

4. Fixed Carbon (%): % FC = 100 - [% Moisture + % VM + % Ash]. High FC indicates superior fuel.
* Ultimate Analysis (Chemical elemental determination):
- Carbon and Hydrogen: Determined by combustion in Liebig train; CO2 absorbed in KOH tubes, H2O absorbed in anhydrous CaCl2 tubes.
- Nitrogen: Determined via Kjeldahl digestion with concentrated H2SO4; liberated NH3 titrated against standard acid.
- Sulfur: Precipitated as BaSO4 using Eschka mixture.

>> 4.4 Knocking, Octane and Cetane Ratings

* Knocking in SI (Spark-Ignition Petrol) Engines: Pre-ignition/auto-ignition of unburned fuel-air mixture ahead of flame front, producing sharp metallic pinging, power loss, and cylinder wear.
* Octane Number: Measure of knock resistance in petrol engines. Standard scale: Iso-octane (2,2,4-trimethylpentane, highly branched, knock resistant) assigned 100; n-heptane (straight-chain, knocks violently) assigned 0.
* Cetane Number: Measure of ignition delay in CI (Compression-Ignition Diesel) engines. Standard scale: Cetane (n-hexadecane, short ignition delay) assigned 100; alpha-methylnaphthalene (long ignition delay) assigned 0.

[SOLVED PROBLEM 1: Bomb Calorimeter Experimental Calculation]

Problem: In a Bomb Calorimeter test, 0.85 g of coal burned completely, causing the temperature of 2,200 g of water in a calorimeter (water equivalent w = 450 g) to rise from 24.20 deg C to 26.85 deg C. Cooling correction delta t = 0.05 deg C; fuse wire correction = 12.0 cal; sulfur acid correction = 25.0 cal; cotton thread correction = 5.0 cal. If coal contains 5.0% hydrogen, calculate HCV and LCV in kcal/kg.
Solution Step 1: Effective Temperature Rise (Delta T):
Delta T = (T2 - T1) + delta t = (26.85 - 24.20) + 0.05 = 2.65 + 0.05 = 2.70 deg C.
Solution Step 2: Total Water Equivalent:
W + w = 2,200 g + 450 g = 2,650 g.
Solution Step 3: Total Corrections:
Total Corrections = 12.0 + 25.0 + 5.0 = 42.0 calories.
Solution Step 4: Calculate Gross Calorific Value (HCV):
HCV = [(W + w) * Delta T - Corrections] / m
HCV = [(2,650 * 2.70) - 42.0] / 0.85 = [7,155 - 42.0] / 0.85 = 7,113 / 0.85 = 8,368.24 cal/g = 8,368.2 kcal/kg.
Solution Step 5: Calculate Net Calorific Value (LCV) with H = 5.0%:
LCV = HCV - [52.83 * %H] = 8,368.2 - [52.83 * 5.0] = 8,368.2 - 264.15 = 8,104.05 kcal/kg.
Final Results: HCV = 8,368.2 kcal/kg | LCV = 8,104.1 kcal/kg.

[SOLVED PROBLEM 2: Dulong's Theoretical Calorific Value Calculation]

Problem: Ultimate analysis of a coal sample yields: Carbon = 82.0%, Hydrogen = 6.0%, Oxygen = 4.0%, Nitrogen = 2.0%, Sulfur = 1.5%, Ash = 4.5%. Calculate HCV and LCV in kcal/kg using Dulong's formula.

Solution Step 1: Available Hydrogen:

Available H = H - (O / 8) = 6.0 - (4.0 / 8) = 6.0 - 0.50 = 5.50%.

Solution Step 2: Apply Dulong's Formula:

HCV = (1 / 100) * [8,080 * C + 34,500 * (H - O/8) + 2,240 * S]

HCV = (1 / 100) * [(8,080 * 82) + (34,500 * 5.50) + (2,240 * 1.5)]

HCV = (1 / 100) * [662,560 + 189,750 + 3,360] = (1 / 100) * [855,670] = 8,556.7 kcal/kg.

Solution Step 3: Calculate LCV:

LCV = HCV - [52.83 * %H] = 8,556.7 - [52.83 * 6.0] = 8,556.7 - 316.98 = 8,239.7 kcal/kg.

Final Answer: HCV = 8,556.7 kcal/kg | LCV = 8,239.7 kcal/kg.

[SOLVED PROBLEM 3: Proximate Analysis Quantification]

Problem: 2.00 g of coal was heated at 105 deg C for 1 hour; residue weighed 1.88 g. The same sample in a covered crucible was heated at 925 deg C for 7 minutes; residue weighed 1.24 g. Burning completely in an open crucible yielded 0.22 g of ash. Calculate proximate analysis percentages.

Solution Step 1: % Moisture:

Loss in weight = 2.00 - 1.88 = 0.12 g.

% Moisture = (0.12 / 2.00) * 100 = 6.0%.

Solution Step 2: % Volatile Matter:

Total loss at 925 deg C = 2.00 - 1.24 = 0.76 g.

Loss due to volatile matter alone = 0.76 - 0.12 (moisture) = 0.64 g.

% Volatile Matter = (0.64 / 2.00) * 100 = 32.0%.

Solution Step 3: % Ash:

Residue = 0.22 g.

% Ash = (0.22 / 2.00) * 100 = 11.0%.

Solution Step 4: % Fixed Carbon:

% Fixed Carbon = 100 - [6.0 + 32.0 + 11.0] = 100 - 49.0 = 51.0%.

Final Report: Moisture = 6.0% | VM = 32.0% | Ash = 11.0% | Fixed Carbon = 51.0%.

[SOLVED PROBLEM 4: Combustion Stoichiometry and Theoretical Air Requirement]

Problem: A fuel gas has the volumetric composition: CH4 = 75%, C2H6 = 15%, CO = 5%, H2 =

3%, N2 = 2%. Calculate the minimum volume of atmospheric air (containing 21% O2 by volume) required for complete combustion of 1 m^3 of this gas.
Solution Step 1: Combustion Chemical Equations per m^3:
1. CH4 + 2 O2 -> CO2 + 2 H2O. O2 required = 0.75 * 2 = 1.50 m^3.
2. C2H6 + 3.5 O2 -> 2 CO2 + 3 H2O. O2 required = 0.15 * 3.5 = 0.525 m^3.
3. CO + 0.5 O2 -> CO2. O2 required = 0.05 * 0.5 = 0.025 m^3.
4. H2 + 0.5 O2 -> H2O. O2 required = 0.03 * 0.5 = 0.015 m^3.
5. N2 is non-combustible.
Solution Step 2: Total Theoretical Oxygen Required:
Total O2 = 1.50 + 0.525 + 0.025 + 0.015 = 2.065 m^3 of pure O2 per m^3 of gas.
Solution Step 3: Calculate Minimum Air Volume (Air is 21% O2 by volume):
Volume of Air = Total O2 * (100 / 21) = 2.065 * 4.762 = 9.833 m^3.
Final Answer: Exactly 9.833 m^3 of air is required per m^3 of fuel gas.

>> 4.5 University Review & Examination Questions

* Q1: Describe the construction, working, and calorific value calculation of a Bomb Calorimeter with corrections.
Answer: Construction: The bomb calorimeter consists of a high-pressure, corrosion-resistant stainless steel vessel (bomb) with a gas-tight screw lid, an inlet valve for oxygen, and two electrical terminals connected to a platinum resistance fuse wire. The fuel pellet (~1 g) sits in a silica crucible. The bomb is submerged in a copper calorimeter containing weighed water (W) and an electrical stirrer, enclosed within an air jacket and water jacket to prevent convective heat losses. Working: Oxygen is filled to 25-30 atm. Fuel is ignited electrically. Water is continuously stirred, and temperature is recorded using a Beckman thermometer readable to 0.001 deg C until peak maximum is observed. Calculations: GCV = [(W + w)*(T2 - T1 + delta t) - (c_fuse + c_acid + c_cotton)] / m. Corrections: (1) Cooling correction delta t: accounts for radiation cooling; (2) Acid correction: exothermic formation of HNO3 and H2SO4 subtracted; (3) Fuse wire correction: heat released by burning fuse wire subtracted.
* Q2: Differentiate between Proximate and Ultimate analysis of coal and explain their industrial significance.
Answer: Proximate analysis determines empirical physical constituents (Moisture, Volatile Matter, Ash, Fixed Carbon) using simple fractional heating regimes. Ultimate analysis determines elemental chemical composition (Carbon, Hydrogen, Nitrogen, Sulfur, Oxygen) via stoichiometry. Industrial Significance: (1) Moisture: Lowers calorific value and consumes transport costs; preferred <2%. (2) Volatile Matter: High VM fuels ignite readily and burn with a long flame, but generate excessive smoke and soot; low VM preferred for metallurgical coke. (3) Ash: Non-combustible inorganic residue; high ash reduces heating value, causes clinkering on boiler grates, and increases disposal costs. (4) Fixed Carbon: Real heat-producing combustible core; higher FC corresponds directly to superior calorific value and boiler thermal efficiency.

>> 4.6 Key Takeaways and Revision Points

* Higher Calorific Value captures latent heat of steam; LCV = HCV - 52.83 * %H kcal/kg.
* Bomb calorimeter measures GCV directly; Dulong’s formula computes GCV from elemental analysis.
* Proximate analysis measures Moisture, VM, Ash, and Fixed Carbon.
* Octane rating measures petrol knock resistance; Cetane rating measures diesel ignition quality.